# Testing Lab: File Management

## Overview

This testing lab covers core file management operations including upload, download, rename, move, delete, and restore from trash.

## Test Environment Setup

### Prerequisites

- FileFlow running on `http://localhost:3000`
- User logged in with write permissions
- Test files prepared:
  - Small text file (10KB)
  - Medium image (2MB)
  - Large video (100MB)
  - Directory structure for move/copy tests

### Test Files Location

- `/tmp/test_files/` (local test directory)
- `/files/test/` (FileFlow directory)

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## Feature 1: File Upload

### Test 1.1: Single File Upload

**Objective**: Verify single file upload functionality
**Steps**: 1. Navigate to file manager 2. Click “Upload” button or drag file to drop zone 3. Select `test_file.txt` (10KB) 4. Verify upload progress bar 5. Wait for upload to complete

**Expected Results**: - ✓ Upload progress bar displayed - ✓ File appears in directory after upload - ✓ File size correct (10KB) - ✓ File modification time correct - ✓ Success notification shown - ✓ File accessible for download immediately

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### Test 1.2: Multiple File Upload

**Objective**: Verify uploading multiple files simultaneously
**Steps**: 1. Click “Upload” button 2. Select 5 files at once 3. Watch upload progress 4. All files upload

**Expected Results**: - ✓ All 5 files uploaded - ✓ Progress shown for each file - ✓ All files appear in directory - ✓ Correct sizes for each file - ✓ No file corruption - ✓ Success message shows count: “5 files uploaded”

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### Test 1.3: Large File Upload (100MB+)

**Objective**: Verify handling of large files
**Steps**: 1. Prepare 150MB video file 2. Click “Upload” 3. Select large file 4. Monitor upload (may take 1-5 minutes) 5. Verify completion

**Expected Results**: - ✓ Upload starts and shows progress - ✓ Progress bar updates smoothly - ✓ Upload completes successfully - ✓ File size correct (150MB) - ✓ File checksum verifiable - ✓ Can download file without corruption

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#### Test 1.4: Drag and Drop Upload

**Objective**: Verify drag-and-drop upload works
**Steps**: 1. Open file manager in one window 2. Open file explorer in another 3. Drag file from explorer to FileFlow 4. Drop on upload area

**Expected Results**: - ✓ Drop zone highlights during drag - ✓ File uploads when dropped - ✓ Upload progress shown - ✓ File appears in directory - ✓ Same behavior as button upload

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#### Test 1.5: Upload to Nested Directory

**Objective**: Verify uploading to subdirectories
**Steps**: 1. Navigate to `/files/test/subfolder1/` 2. Upload a file 3. Verify file appears in correct location

**Expected Results**: - ✓ File uploads to correct directory - ✓ Full path preserved - ✓ File visible in subfolder - ✓ Breadcrumb shows correct location

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#### Test 1.6: Duplicate Filename Upload

**Objective**: Verify handling of duplicate filenames
**Steps**: 1. Upload `document.txt` (first time) 2. Upload `document.txt` again

**Expected Results**: - ✓ First upload succeeds - ✓ Second upload shows conflict dialog - ✓ Options: "Replace" | "Keep Both" | "Cancel" - ✓ If "Keep Both": file renamed to `document (1).txt` - ✓ If "Replace": original overwritten - ✓ Both options work correctly

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#### Test 1.7: Filename Sanitization

**Objective**: Verify dangerous characters removed from filenames
**Steps**: 1. Create file named: `../../../etc/passwd` 2. Upload the file 3. Check resulting filename in FileFlow

**Expected Results**: - ✓ Path traversal characters removed - ✓ File uploaded safely - ✓ Filename sanitized: `etc_passwd` (or similar) - ✓ File stored in upload directory only - ✓ No system files accessed or modified

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### Feature 2: File Download

#### Test 2.1: Single File Download

**Objective**: Verify downloading individual files
**Steps**: 1. Navigate to file manager 2. Right-click on file 3. Select "Download" 4. File downloads to local machine

**Expected Results**: - ✓ Download starts immediately - ✓ File downloaded to Downloads folder - ✓ Filename matches original - ✓ File size correct - ✓ File integrity maintained - ✓ Can open and read file

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#### Test 2.2: Bulk Download as ZIP

**Objective**: Verify downloading multiple files as archive
**Steps**: 1. Select 5 files 2. Right-click 3. Select "Download as ZIP" 4. ZIP downloads

**Expected Results**: - ✓ ZIP file created - ✓ Contains all 5 files - ✓ Folder structure preserved - ✓ ZIP size reasonable - ✓ ZIP can be extracted - ✓ All files intact after extraction

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#### Test 2.3: Download Large File

**Objective**: Verify downloading files >100MB
**Steps**: 1. Download 150MB video file 2. Monitor download progress 3. Verify file integrity

**Expected Results**: - ✓ Download starts - ✓ Progress shown - ✓ Download completes - ✓ File size correct (150MB) - ✓ MD5/SHA checksum matches - ✓ File can be played/opened

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#### Test 2.4: Download Interrupted and Resumed

**Objective**: Verify resume capability if download interrupted
**Steps**: 1. Start downloading large file 2. Pause download after 50% complete 3. Resume download 4. Verify completion

**Expected Results**: - ✓ Resume feature available - ✓ Continues from paused point - ✓ No need to re-download from start - ✓ Final file complete and valid

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### Feature 3: File Rename

#### Test 3.1: Rename File

**Objective**: Verify file rename functionality
**Steps**: 1. Right-click on file: `old_name.txt` 2. Select "Rename" 3. Enter new name: `new_name.txt` 4. Press Enter or click "Save"

**Expected Results**: - ✓ File renamed successfully - ✓ New name appears in directory - ✓ Old name no longer visible - ✓ File size and content unchanged - ✓ Modification timestamp updated - ✓ File still accessible with new name

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#### Test 3.2: Rename with Special Characters

**Objective**: Verify handling of special characters
**Steps**: 1. Rename file to: `document (2024-05-05) [draft].txt` 2. Verify rename succeeds

**Expected Results**: - ✓ File renamed with allowed special chars - ✓ Parentheses, brackets, hyphens preserved - ✓ File accessible with new name - ✓ Special characters don't cause issues

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#### Test 3.3: Rename to Existing Name

**Objective**: Verify handling duplicate names
**Steps**: 1. Have files: `file1.txt` and `file2.txt` 2. Try to rename `file2.txt` to `file1.txt`

**Expected Results**: - ✓ Rename prevented - ✓ Error message: "File already exists" - ✓ Original filename retained - ✓ User can choose different name

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#### Test 3.4: Rename Directory

**Objective**: Verify renaming folders
**Steps**: 1. Right-click on folder: `oldname/` 2. Select "Rename" 3. Enter new name: `newname` 4. Confirm

**Expected Results**: - ✓ Folder renamed - ✓ All contents preserved - ✓ Paths updated correctly - ✓ Files within still accessible - ✓ Internal shares, shared folders, favorites, public share links and upload requests on the folder (or anything inside it) now point to the new path and keep working

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### Feature 4: Move and Copy

#### Test 4.1: Move File to Different Directory

**Objective**: Verify moving files between directories
**Steps**: 1. Right-click on file in `/files/test/` 2. Select "Cut" 3. Navigate to `/files/archive/` 4. Right-click and select "Paste"

**Expected Results**: - ✓ File moves to new location - ✓ Original location shows file gone - ✓ New location shows file - ✓ File size unchanged - ✓ File content intact - ✓ Modification time updated - ✓ An existing public share link to the file still downloads it from its new location

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#### Test 4.2: Copy File

**Objective**: Verify copying files
**Steps**: 1. Right-click on file 2. Select "Copy" 3. Navigate to destination folder 4. Select "Paste"

**Expected Results**: - ✓ File copied to new location - ✓ Original file remains - ✓ Copy has same name - ✓ Copy has identical content - ✓ Copy has new creation timestamp - ✓ Both files independent

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#### Test 4.3: Move Multiple Files

**Objective**: Verify moving multiple files at once
**Steps**: 1. Select 5 files 2. Right-click "Cut" 3. Navigate to destination 4. "Paste"

**Expected Results**: - ✓ All 5 files move together - ✓ No files left behind - ✓ All appear in destination - ✓ Batch operation succeeds

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#### Test 4.4: Move Directory with Contents

**Objective**: Verify moving folders with files
**Steps**: 1. Have directory with 10+ files 2. Move entire directory to new location 3. Verify all contents moved

**Expected Results**: - ✓ Entire directory structure preserved - ✓ All nested files moved - ✓ No files lost - ✓ Directory accessible at new path - ✓ Public links to files inside the directory and upload requests targeting it follow the move

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## Feature 5: Delete and Trash

### Test 5.1: Delete File to Trash

**Objective**: Verify files go to trash, not permanent delete
**Steps**: 1. Right-click on file 2. Select "Delete" or press Delete key 3. Confirm deletion 4. Check Trash folder

**Expected Results**: - ✓ File no longer in original location - ✓ File appears in Trash - ✓ Trash shows deleted timestamp - ✓ Can restore from Trash - ✓ Original path remembered - ✓ Public share links and upload requests on the file (or on anything inside a deleted folder) are removed: their URLs return 404, also after a restore

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### Test 5.2: Permanent Delete from Trash

**Objective**: Verify files can be permanently deleted
**Steps**: 1. Navigate to Trash 2. Select file 3. Click "Delete Permanently" 4. Confirm

**Expected Results**: - ✓ File removed from Trash - ✓ File no longer recoverable - ✓ Storage space freed - ✓ File disappears immediately

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### Test 5.3: Restore from Trash

**Objective**: Verify files can be restored
**Steps**: 1. Delete file to Trash 2. Navigate to Trash 3. Click "Restore" on file 4. Verify file back in original location

**Expected Results**: - ✓ File restored to original directory - ✓ File content unchanged - ✓ File timestamps preserved - ✓ File not in Trash anymore

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### Test 5.4: Auto-Empty Trash After 30 Days

**Objective**: Verify auto-deletion of old trash items
**Steps**: 1. Delete file to Trash 2. Set system clock forward 30 days 3. Check Trash

**Expected Results**: - ✓ Files older than 30 days auto-deleted - ✓ Newer files retained - ✓ No manual cleanup needed - ✓ Storage freed automatically

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### Test 5.5: Links Do Not Attach to a Recreated File

**Objective**: Verify a public link cannot silently expose a new file created at the same path
**Steps**: 1. Create a public share link for `/files/test/report.txt` 2. Delete `report.txt` 3. Upload a different file with the same name to `/files/test/` 4. Open the old share link

**Expected Results**: - ✓ The old link returns 404 ("not found or expired") - ✓ The new file is not exposed; a new link must be created for it

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## Performance Tests

### Test 6.1: Upload 100 Small Files

**Objective**: Verify handling of many files
**Steps**: 1. Create 100 text files (1KB each) 2. Upload all files 3. Verify all appear in directory 4. List directory

**Expected Results**: - ✓ All 100 files upload - ✓ Directory listing shows all files - ✓ No performance degradation - ✓ Can still navigate and search

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### Test 6.2: Bulk Operation on 500+ Files

**Objective**: Verify bulk operations scale
**Steps**: 1. Have directory with 500+ files 2. Select all files 3. Apply bulk tag or bulk move 4. Verify completion

**Expected Results**: - ✓ Operation completes in <10 seconds - ✓ All files affected correctly - ✓ No timeouts - ✓ UI remains responsive

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## Security Tests

### Test 7.1: Dot-Segment Paths Rejected

**Objective**: Verify `..` segments cannot escape a group's allowed paths or a shared folder
**Steps**: 1. Login as a user whose group is restricted to `/teamA` 2. Call `GET /api/files/download?path=/teamA/..` 3. Call `GET /api/files?path=/teamA/./sub//` (listing with `.` and doubled slashes)

**Expected Results**: - ✓ Step 2 rejected with HTTP 400 ("Invalid path"); no content from `/teamB` returned - ✓ Step 3 treated as `/teamA/sub` (normalized), allowed as usual - ✓ A direct request for `/teamB/secret.txt` returns 403

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## Test Results Summary

| Test ID | Test Name | Status | Notes |
|---|---|---|---|
| 1.1 | Single Upload | ☐ | |
| 1.2 | Multiple Upload | ☐ | |
| 1.3 | Large File Upload | ☐ | |
| 1.4 | Drag & Drop | ☐ | |
| 1.5 | Upload to Subfolder | ☐ | |
| 1.6 | Duplicate Filename | ☐ | |
| 1.7 | Filename Sanitization | ☐ | |
| 2.1 | Single Download | ☐ | |
| 2.2 | Bulk ZIP Download | ☐ | |
| 2.3 | Large Download | ☐ | |
| 2.4 | Download Resume | ☐ | |
| 3.1 | Rename File | ☐ | |
| 3.2 | Special Characters | ☐ | |
| 3.3 | Duplicate Name | ☐ | |
| 3.4 | Rename Directory | ☐ | |
| 4.1 | Move File | ☐ | |
| 4.2 | Copy File | ☐ | |

| Test ID | Test Name | Status | Notes |
|---|---|---|---|
| 4.3 | Move Multiple | ☐ | |
| 4.4 | Move Directory | ☐ | |
| 5.1 | Delete to Trash | ☐ | |
| 5.2 | Permanent Delete | ☐ | |
| 5.3 | Restore from Trash | ☐ | |
| 5.4 | Auto-Empty Trash | ☐ | |
| 5.5 | Links vs Recreated File | ☐ | |
| 6.1 | 100 File Upload | ☐ | |
| 6.2 | 500+ Files Bulk Op | ☐ | |
| 7.1 | Dot-Segment Paths | ☐ | |

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## Sign-Off

- **Tester Name**: _____
- **Test Date**: _____
- **Total Tests**: 27
- **Passed**: ____
- **Failed**: ____
- **Blocked**: ____

**Conclusion**: ☐ Ready for Production | ☐ Needs Fixes | ☐ Blocked

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## Version History

| Date | Version | Changes | Author |
|---|---|---|---|
| 2024-05-05 | 1.0 | Initial lab | Driss OUAAZIZ |
| 2026-10-27 | 1.1 | Public links and upload requests follow rename/move and are removed on delete; tests 5.5 and 7.1 (dot-segment paths) added | Driss OUAAZIZ |